

ASSIGNMENT NO 2

Topic: Command Line Calculator

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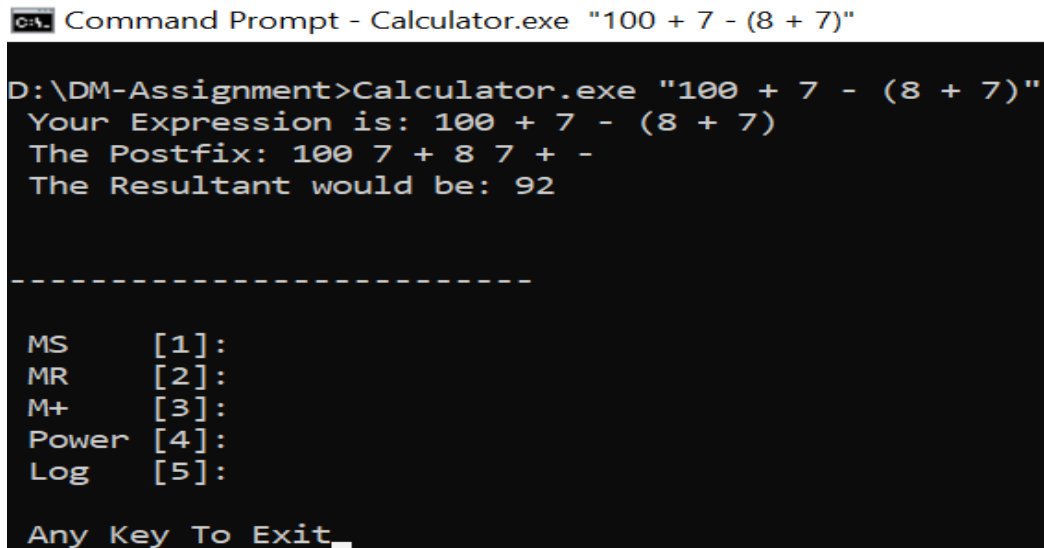
Using Through Command Line

When a User wants to use my Command Line Calculator program. He/She just needs to pass the following arguments

- 1- Name of the File Calculator.exe
- 2- Then he can write the Equation in the inverted commas and press enter. User must add spaces between the operator and operands For example:

Calculator.exe "100 + 7 - (8 + 7)"

He will get its postfix form and resultant mathematical solution.



```
Command Prompt - Calculator.exe "100 + 7 - (8 + 7)"

D:\DM-Assignment>Calculator.exe "100 + 7 - (8 + 7)"
Your Expression is: 100 + 7 - (8 + 7)
The Postfix: 100 7 + 8 7 + -
The Resultant would be: 92

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MS      [1]:
MR      [2]:
M+      [3]:
Power   [4]:
Log     [5]:

Any Key To Exit_
```

He will get a list of other options such as power function and log function.

MS is used to store the current result.

MR is used to display the stored results in the memory.

M+ add the previous stored results with the current result and stores them back in the memory.

3- Program also check for validity as well if the user tries to input wrong equation it will not display results. Such Errors Include:

- a) Writing Equation without spaces between operators and operands.
- b) Writing Equation with non numeric operands like A, B or \$ etc.
- c) Writing Equation with operands other than (+, -, * , /)
- d) Writing Operators more than the number of operands.
- e) Writing equation with imbalanced brackets.

```
D:\DM-Assignment>Calculator.exe 100 + 9 - (( 9 -8 )
Brackets not Balanced!
Please Input Valid Expression!
D:\DM-Assignment>
```

 Command Prompt

```
D:\DM-Assignment>Calculator.exe 100 + 99 - 932 + 333 -
Your Expression is: 100 + 99 - 932 + 333 -
The Postfix: 100 99 + 932 - 333 + -
Cant Evaluated Expression Because Operators or Operands are not correct!
D:\DM-Assignment>_
```

Here no of operators are not correct so simplified expression can't be calculated!

```
D:\DM-Assignment>Calculator.exe "100 + A (* (0))"  
Brackets not Balanced!  
Please Input Valid Expression!  
D:\DM-Assignment>_
```

Note:

User can add multiple Arguments and my program is designed to handle it as one argument.

If User does not Enter any Command

Then My program will ask the user for input!

```
D:\DM-Assignment>Calculator.exe  
Please Input Expression!:
```

Here no Argument is given except Calculator.exe so program ask for user input.

When user enters input he must input without Inverted Commas and must add spaces after operators and operands.